

Flexible and adaptive coherent PON for next-generation optical access network [Invited]

Junwen Zhang^{*}, Guoqiang Li, Sizhe Xing, Nan Chi

^a Department of Communication Science and Engineering, Key Laboratory for Information Science of Electromagnetic Waves (MoE), Fudan University, Shanghai, China

^b Peng Cheng Laboratory, Shenzhen 518055, China

ABSTRACT

With the official adoption of the single-wavelength 50G PON standard at the International Telecommunication Union Study Group 15 (ITU-T SG15) meeting, both industry and academia have started extensive research on the next-generation optical access network beyond 50G. It is envisioned that 100G and 200G per wavelength and even beyond will be required in the next-generation optical access network to support the continuously growing bandwidth demands. Coherent passive optical network (C-PON), which is based on multi-access coherent optics and digital signal processing (DSP), appears to be a promising candidate for future 100/200G and even beyond single-wavelength PON, thanks to its superior sensitivity compared with current direct-detection based PONs. Actually, multi-access coherent optics brings many benefits to the optical access network beyond the improvement of sensitivity. In this invited paper, we review recent studies and progress on the flexible C-PON for optical access network in terms of access speed, multiple-access architecture, and deployment scenarios. Rate-adaptive coherent access is first introduced in this paper as a promising solution to achieve flexible C-PON. Adaptive modulation formats based on constellation shaping is highlighted to realize fine granularity of information rate and wide flexibility. We have also discussed the flexible multiplexing schemes beyond TDM, to take the full advantages of coherent access. Finally, we analyze the flexible deployment scenarios under the large power budgets thanks to coherent detection. All these flexibilities provided by coherent access would be interesting and beneficial in the next generation optical access network.

1. Introduction

The bandwidth demand in optical access network has grown tremendously in recent years, thanks to the requirements from emerging services such as 8 K/16 K high-quality video streaming applications, Virtual Reality/Augmented Reality (VR/AR), 5G mobile Internet, and edge computing services [1–4]. Passive optical network (PON) as the widely deployed low-cost solution for “the first mile” connection between users and service providers, has been developed with several generations in the last two decades [1]. Fig. 1 shows the PON standards development in two major industry standardization bodies, i.e., the IEEE 802.3 and the International Telecommunication Union Study Group 15 (ITU-T SG15) collaborating with Full Service Access Network (FSAN) Group. Two types of PONs, namely EPON and GPON, have been developed. These different types of PONs can be grouped as three major generations, which are shown in Fig. 1, as the Gbps-PON era in 2000 to 2005, the 10Gbps-PON era around 2010 to 2015, and the latest 25/50Gbps-PON era around 2020. Moving forward, it is expected that 100G and 200G per wavelength and even beyond will be required in the next-generation optical access network to support the continuously growing bandwidth demand towards 2030 [3,4]. Many applications,

from residential, to mobile X-haul, and business services, are envisioned to require aggregated access bandwidth up to 100/200G [4]. Take mobile X-haul in B5G/6G for example, up to 86.1Gbps per link will be required for a 100 MHz radio frequency bandwidth, 256-QAM (quadrature amplitude modulation), 8 MIMO (multiple input multiple output) layers, and 32 antenna ports with Fx interface [2]. It can be even higher for future 6G system with larger radio frequency bandwidth when considering millimeter-wave (mmWave) frequencies towards 2030 [2,5].

The ever-increasing demand for bandwidth is one of the most important factors driving the development of high-speed PON [3]. Current PON systems are all based on intensity modulation and direct-detection (IM/DD); unfortunately, the sensitivity of single-wavelength IM/DD PON is almost reaching its limit at 50G considering the required power budget. The receiver sensitivity is defined at -22.75 dBm in the ITU-T 50G PON standard (G.9804) at the BER level of 1×10^{-2} [6]. Going up to 100G and 200G, it will be very challenging to maintain PR-30 power budget of 29 dB for IM/DD solution. Considering practical applications, coexistence with legacy PON systems in operation requires a power budget of 31 dB or even higher. Indeed, most recently, people are getting better power budgets in for IM/DD PONs by using advanced

^{*} Corresponding author.

E-mail address: junwenzhang@fudan.edu.cn (J. Zhang).

signal processing and high-launch power, i.e., 14 dBm is required to achieve the 34 dB power budget in [7]. However, it is still very challenging to achieve such a high output power in the optical network unit (ONU) side for upstream. Additionally, O-band wavelength is required to avoid the fiber dispersion penalty for IM/DD PONs. This makes it more challenging to consider co-existing with legacy PON systems which are all in O-band. Therefore, superior sensitivity is one of the most significant motivations of introducing coherent optics into optical access network, thanks to the coherent beating gain from the local oscillator (LO). For example, a sensitivity of -38 dBm at the BER of 1×10^{-3} has been demonstrated for 100G polarization-division-multiplexing Quadrature Phase Shift Keying (PDM-QPSK) single-wavelength TDM-PON [8]. It can provide extended power budget for long-reach and large-split-ratio access in next-generation optical access network.

The multi-access coherent optics that is based on coherent detection for optical access network has received extensive research interests in PON most recently thanks to its unique advantages compared with IM/DD solutions [4]. Besides the advantage of superior receiver sensitivity, multi-access coherent optics also have the benefits of high access speed achieved by using advanced modulation formats, and digital dispersion compensation as well as equalizations enabled by digital signal process (DSP) thanks to the linear conversion of optical field to digital domain. C-band wavelength resources and mature optoelectronic components can be used without occupying wavelengths in the already crowded O-band window. Therefore, C-PON can easily co-exists with a legacy PON system in the same optical distribution network (ODN). Another significant unique advantage of coherent optics is the inherent frequency-selective capability by coherent detection. This will be beneficial to flexible frequency-division multiplexing without using any optical filters. All of these unique advantages provided by coherent optics makes it an attractive solution for the multi-access scenario in optical access network. Recent demonstrations of coherent PON from 100G to near 1Tbps have shown promising results in terms of power budget, dynamic range and flexibility [9].

Flexibility is becoming increasingly attractive, and it may be one of the most interesting features of next-generation access networks [10–13]. As the application scenarios of PON have been greatly extended beyond residential services, thanks to the emerging services such as mobile X-haul, edge-computing and so on [14]. The deployment scenarios have also been extended to support different use cases, from FTTH (fiber to the home), to FTTO (office), FTTB (building) and FTTC (mobile cell) [4]. Current PON systems provide all users with the same access data rate, namely the guaranteed data rate, which is unfortunately limited by the worst performing Optical Network Unit (ONU). However, since a PON system is naturally a point-to-multipoint (PtMP) architecture, it is ideal for flexible access data rates. To solve this problem, flexible PON or namely rate-adaptive PON has recently been studied based on either adaptive coding rate or adaptive modulation formats in both IM/DD PON and C-PON [10–13]. With much larger

dynamic range and power budget, flexible C-PON can achieve higher optimized capacity [10,12].

In this paper, we review the recent advances of flexible C-PON in terms of access speed, architecture and deployment scenarios. We first introduce the rate-adaptive coherent PON as a promising solution to achieve flexible C-PON. Different schemes to achieve adaptive coherent access speed are analyzed. Adaptive modulation formats based on constellation shaping is highlighted to realize fine granularity of information rate and wide flexibility, with the demonstrations of 100G to 300G rate-adaptive flexible C-PON [10]. We have also discussed the flexible multiplexing schemes beyond TDM, to take the full advantages of coherent access. A 100G single-wavelength TFDm flexible PON are introduced and highlighted [15]. Finally, we analyze the flexible deployment scenarios under the large power budgets thanks to coherent detection [16]. We believe all these flexibilities provided by coherent access would be interesting and beneficial in the next generation optical access network. A portion of the experimental results have been presented in the following references [10,15,17,18], in which the details can be found.

The remainder of the paper is organized as follows. In section 2, we describe the principle and demonstrations of rate-adaptive C-PON. Then we introduce the flexible multiplexing schemes beyond TDM in section 3. The flexible deployment scenarios in C-PON enabled by large power budgets is discussed in section 4. Finally, section 6 concludes the paper.

2. Rate-adaptive access for flexible C-PON

2.1. Motivations and realization

As mentioned above in section 1, there are increasing interests in exploring the adaptive access data rates in the flexible PON [4,19]. In both IM/DD PON and C-PON, many studies have been reported to achieve adaptive line access rates for different users or different groups of users based on their channel conditions, i.e., the optical path loss (OPL) variations [10,12,19,20]. As shown in Fig. 2, if three groups of users are located in different locations with different OPL, the access rate that a group of users can achieve can be optimized based on their channel conditions. For example, if $OPL1 < OPL2 < OPL3$, then the maximum achievable data rates $Rate1 > Rate2 > Rate3$, accordingly. Today's fixed data rates in legacy PON systems that is determined by the ONU with the worst channel condition limits the overall access throughput. It is worth noting that the idea adaptive rates for flexible access has already been applied in other wireless and wired access network, such as Wi-Fi, mobile network and cable-based networks [13,21]. In these systems, adaptive modulation formats are used based on their channel conditions to achieve reliable and high-capacity communications. PON is naturally a quasi-static distributed network with different users at different locations, which means some of them are with very good channel conditions. Higher-orders of modulation formats can

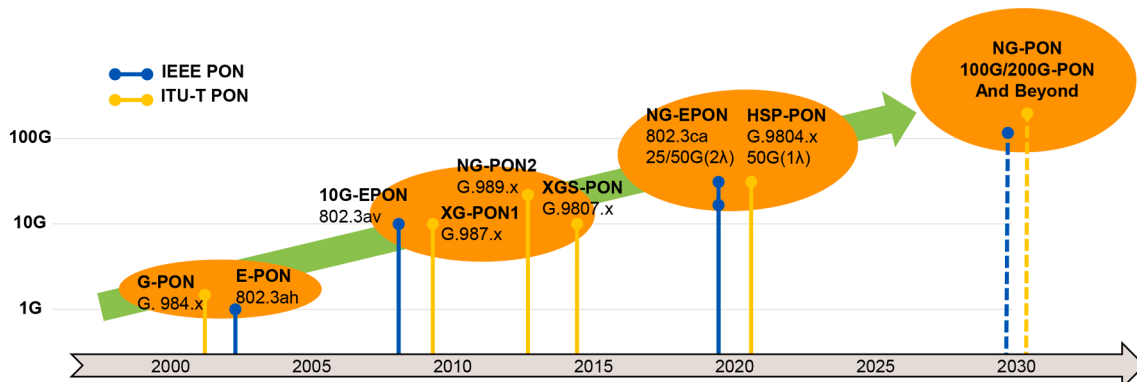


Fig. 1. The PON evolution with different generations and the timeline for future 100G/200G and beyond next-generation PON.

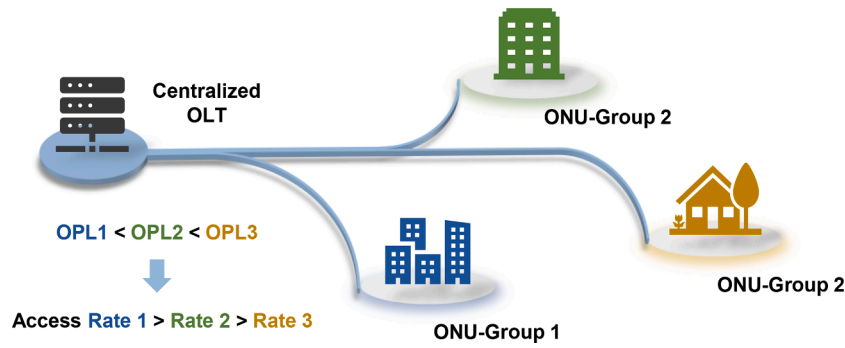


Fig. 2. The motivation and principle of rate-adaptive access for flexible PON.

be assigned to these users to achieve higher data rates; therefore, it is of great interest for operators to implement flexible PON with adaptive access data rates in their network. The performances are improved at low-cost and overall network access capacity is also improved. Most importantly, generally the adaptive access is compatible with the TDM access architecture without changing to its multiplexing scheme.

Many rate-adaptive IM-DD PON has been reported based on different schemes, including multi-rate NRZ modulation, adaptive modulation formats (e.g., from NRZ to PAM4 [22]), and adaptive coding schemes (e.g., different forward-error-correction (FEC) code-rates [13,19]). In [23], it shows a fine granularity scheme based on the bits allocation of DMT modulation. The rate of the ONU can be adjusted by changing the target total bits with a granularity of one bit. The proposed flexible-rate PON can achieve the widest-range rate adjustment from 25 Gb/s to 100 Gb/s with a granularity of ~ 50 Mbit/s. Compared with IM-DD PON in which signals are modulated on single dimension, namely, the amplitude or intensity, coherent optics employ multiple dimensions for signal modulation, including phase, amplitude, polarization and so on. In addition, coherent optics generally have the advantages of higher data rates and improved dynamic range compared with IM/DD. Therefore, more degrees of freedom can be explored in coherent PON to achieve rate adaptive access. Table 1 summarizes different schemes for rate-adaptive C-PON, including the key features and the information rate change granularity of these schemes.

The adaptive rates can be simply realized by changing different signal baud rates, as demonstrated in [24], where two different baud rates are supported to achieved flexible data-rates. It requires the

transceivers to work at different clock and sampling rates. Alternative, a rate-adaptive C-PON can be realized by adjusting the order of the modulation formats with the same baud rate. Vacondio *et al.* demonstrated a flexible TDM optical access network based on software-defined coherent transponders capable of exchanging optical data bursts with payloads at different data-rates [11]. The adaptive access data-rates are depending on the link power budget, with modulation formats ranging from polarization-division multiplexed binary phase shift keying (PDM-BPSK) QPSK, 8QAM to 16-QAM. Results show an increased average capacity per user up to a factor more than 2 can be achieved under some OLT-ONU distances distributions [11]. To further explore the flexibility in coherent optics, a multidimensional data-rate and power-budget adaptive coherent PON can be realized from the in-phase/quadrature (I/Q) channels, and orthogonal polarizations of complex modulation formats. As demonstrated in [25], a rate-adaptive I/Q-imbalanced modulation is demonstrated in the PDM-QPSK 100G C-PON.

However, the data-rates tuning based on above mentioned uniform distributed modulation formats is generally stepwise, and there are large sensitivity differences between different orders of signals. An effective solution is to use flexible FEC code rates to reduce the information rate gap between different modulation formats. For example, Mu *et al.* demonstrated a rate-adaptive C-PON by using adaptive modulation-and-coding (MCS) scheme in [12], where 12 MCS options can be achieved by using four different code rates with three modulation formats ranging from PDM-QPSK, 16QAM to 64 QAM. The gaps of net data-rates are reduced by using different FEC code rates.

Constellation shaping and design is an alternative scheme to explore in rate-adaptive coherent access. For example, probabilistic shaping (PS) can provide a fine granularity of net data rate [10]. It has been widely used in the long-haul coherent communication systems and can achieve wider flexibility with respect to the flexible FEC scheme [12,26]. By changing the information source entropy according to the channel conditions, PS can approach the optimal channel capacity with fine granularity. As demonstrated in our recent work [10], a 300G rate-adaptive C-PON is achieved by PS with net data rate varying from 255-Gbits/s to 85-Gbits/s as the optical path loss varies from 3-dB to 38-dB. The details of this work will be presented in the next section.

2.2. Demonstration of 300G flexible C-PON based on PS-QAM

In this section, we introduce our most recent work on the rate-flexible TDM C-PON in upstream burst-mode detection based on PS-QAM modulation [10]. In this work, we apply the local-oscillator (LO) power adjustment in our proposed work to enable large dynamic range [18], achieving 300G peak line access data rate with 25-GBaud QAMs and a dynamic range of over 35 dB [10]. Besides the PS-QAM modulation, burst-mode frame generation and burst-mode signal processing [8], is also essential to enable the TDM access in this demonstration.

Fig. 2 shows the schematic of the experimental demonstration as well as the principles of PS-QAM signal generation, burst-frame structure, and receiver-side burst-mode signal processing. The detailed DSP of

Table 1
Rate-adaptive C-PON schemes.

Scheme	Key features	Peak Rate (Gbps)	Granularity	Reference
Adaptive Baud Rate	Two different baud rates to support different data rates	100	Large	[24]
Adaptive Modulation Orders	Four modulation formats from BPSK to 16QAM	100	Large	[11]
Multidimensional Modulation	Rate-adaptive by I/Q-imbalanced modulation	100	Large	[25]
Adaptive Modulation and Coding	12 MCS options using four code rates with three modulation formats	384	Medium	[12]
Adaptive Probabilistic Shaping	Adaptive source entropy according to the channel conditions	300	Fine	[10]

burst-mode signal generation with PS-QAM mapping and synchronization pattern (SP) insertion is shown Fig. 2(a). Fig. 2(b) shows the detailed process of burst frame generation with probabilistically amplitude shaping and FEC encoding [27]. Here we take 64QAM as an example, the raw data bits are first separated into two branches of U_1 (Data 1) and U_2 bits (Data 2). Then, to achieve QAM constellation distribution with Maxwell-Boltzmann distribution, we apply the probabilistically amplitude shaping process based on the U_2 bits to generate 2 V bits. This process is realized in the distribution matching (DM) part in Fig. 2(b). After that, the Data 1 and shaped Data 2 is combined to generate the FEC head after the FEC encoding process. The V symbols of PS-QAM is generated after QAM-mapping process with both I and Q paths. Finally, the burst frame can be generated by properly adding the preamble with the payload [8]. To optimize the overall capacity to different groups of ONUs based on the available channel conditions, i.e., the OPL and the associated SNR and nonlinearity (NL) impairments, the signals undergoes a series pre-process under the instruct of the channel estimation as shown in Fig. 2(c). Here, the source entropy of PS-QAM is determined by the results of channel estimation. After mapping to PS-QAM signals, we apply look-up-table (LUT) technology to pre-equalize the distortion according to the channel conditions [10].

The receiver-side DSP is shown in Fig. 2 (d) and the details of the burst-mode DSP is shown in Fig. 2 (e). The detailed burst-mode DSP can be found in [8], where QPSK signals are equalized and demodulated in burst-mode. Same algorithms can be applied to the PS-QAM signals. Fig. 3 shows the final experimental results of rate-adaptive 300G TDM C-PON based on PS-QAM in upstream, with net data rate varies from 255-Gbits/s to 85-Gbits/s as the optical path loss varies from 3-dB to 38-dB after a 20-km SSMF transmission [10]. The constellations of (i) to (iv) shows the source entropy changes under different OPLs.

3. Flexible multiple-access beyond TDM

3.1. Motivations and realization

In the previous section, rate-adaptive coherent access technology, based on advanced modulation formats, is introduced for flexible TDM C-PON. TDM has dominated the bandwidth sharing scheme in PON for decades [22]. The TDM C-PON saves the wavelength resources and simplify the deployment, due to the advantages of statistical multiplexing for bandwidth sharing and transparent signal transmission with colorless components [10]. However, TDM-PON faces the challenges to support low scheduling latency and high-bandwidth requirements for high-end users. The use of coherent optic in optical access network enables more flexible multiple-access schemes. In this section, we introduce the flexible multiple-access schemes beyond TDM, and show how

we can take the full advantages of coherent optics in access network.

Fig. 4 shows the different multiple access technologies that is possible for coherent optical access network. TDMA is the most studied multiple-access multiplexing mechanism in many C-PON demonstrations [10,11,28,29]. Besides the TDM C-PON, wavelength-division multiplexing (WDM) or ultra-dense WDM (UDWDM) is another strong candidate for multi-access coherent optics, which allocates specified wavelengths to the high-end users [30–32]. The wavelength-division multiple access (WDMA) provides separate, low latency and secure virtual point-to-point services. In a C-PON system, the UDWDM is enabled by the inherent frequency selectivity of coherent optics in the same split-based ODN without any optical filters [30]. This unique advantage from coherent optics makes the fine frequency selection and frequency-division multiplexing (FDM) C-PON possible. It enables flexible sharing of the total bandwidth by allocating sub-bands to different end-users based on the digital subcarrier multiplexing technology [33–35]. The digital sub-bands are modulated on one-wavelength in the downstream, and sub-bands from different end-users can be received by a single broadband coherent receiver [35]. Since the data aggregation occurs in the frequency domain, it faces the challenge of signal overlapping, and guard bands are reserved between FDM channels, which will limit the spectral efficiency.

While TDM, FDM and WDM share the available time and spectrum resources, non-orthogonal multiple access (NOMA) is another solution

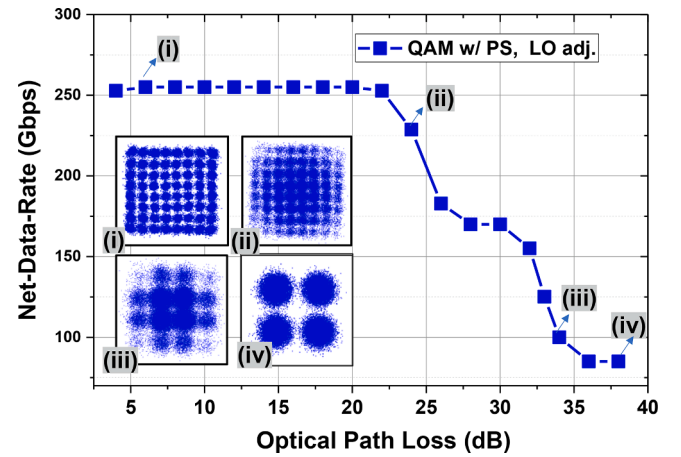


Fig. 4. The experimental results of rate-adaptive 300G coherent PON based on PS-QAM, showing the net data rates under different OPLs. The constellations of (i) to (iv) shows the source entropy changes under different OPLs.

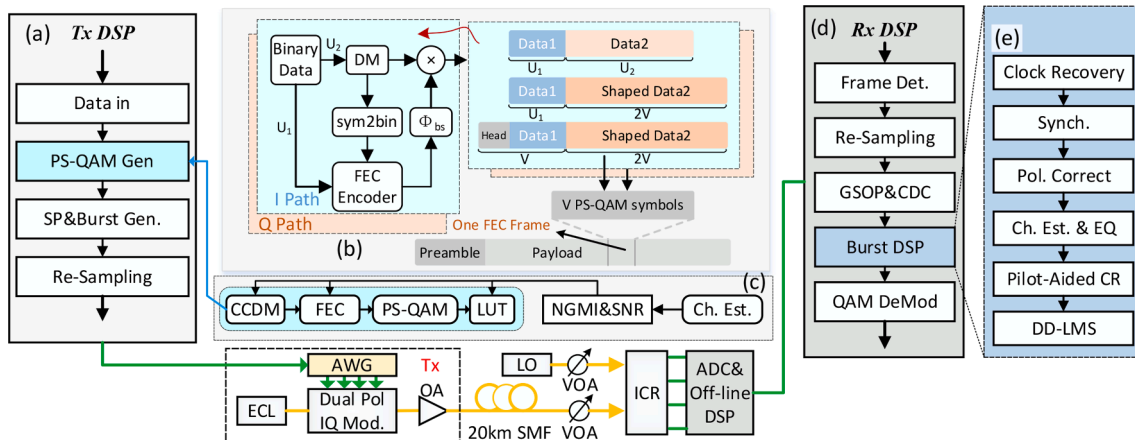


Fig. 3. The schematic of the experimental demonstration of the 300G Flexible CPON, including (a) the Tx-side DSP, (b) the detailed process of burst frame generation based on PS-QAM and FEC encoder, (c) the process of channel estimation, coding, and QAM generation, (d) the Rx-side DSP and (e) the detailed burst-mode DSP.

in power-domain [36]. NOMA enhances the capacity of the network by combining the information of the users by constellation multiplexing [36]. By applying NOMA with C-PON, the reach and splitting factor of the network can be achieved through the increment in optical power budget given by the coherent receiver. Finally, code-division multiplexing (CDM) is also one of the multiple-access technologies that can be applied to coherent optical access network for flexible access by multiplexing users across the same timeslot and spectrum through a specific code [37]. In CDM C-PON system, different ONUs will occupy the same frequency band and timeslot simultaneously without filtering with one another because these ONUs have different spreading sequence. All the signals can be received simultaneously and separated by their own unique despreading sequence.

In addition to these different multiple access schemes, two- or multi-dimensional coherent access will be of great interest for next generation access network, which is enabled by combining TDM, FDM, WDM and the other access technology into one coherent system. For example, WDM-FDM is possible by applying digital subcarriers on multiple optical carriers on different wavelengths to achieve over Tbps access [38]. A 16-Tb/s ultra-high capacity coherent point-to-multipoint network can be realized by coherent DWDM-CDM [37]. NOMA-FDM can be realized by using the pilot-tone based interference cancellation, which enables power- and frequency-division multiplexing (PFDM) coherent optical access [39]. Time-frequency-division multiplexing (TFDM) C-PON is also another example of flexible multiple access that takes the advantages of both TDM and FDM C-PON [15,40]. It enables more flexible bandwidth allocation. In the following part, we will introduce our previous work of a 100G single-wavelength TFDM flexible C-PON [15].

3.2. Example of the flexible TFDM C-PON

Fig. 5(a) shows the architecture of the TFDM coherent PON based on digital subcarrier multiplexing technology, and most recently a flexible single wave-wavelength TFDM 100 Gb/s C-PON was demonstrated in [15]. In this demonstration, the 100G TFDM C-PON system has four subcarriers, each of which transmits 25-Gbps data rate based on PDM-QPSK. The number of sub-bands, the bandwidth of each sub-band as well as the modulation formats can be different and can be extended according to the channel conditions. In the downstream of this demonstration, the four subcarriers are generated on the single wavelength by using an I/Q modulator at the OLT side. The receiver at the ONU side can detect the multi-subcarrier signal separately [15]. In the upstream, each ONU generates one to four subcarriers, which means that the aggregated upstream signal has four subcarriers that may come from the same ONU or different ONUs. At the OLT side, only single broadband coherent receiver is needed for subcarriers detection due to

the advantages of wavelength selection in coherent detection [15]. This is the main difference between IM/DD and coherent detection, as carrier beating interferences makes it impossible for IM/DD. It is worth noting that the transceiver in the OLT and the ONU can be asymmetrical, and the digital sub-bands can be shared by more numbers of ONUs and TDM can be applied in this case.

The possible bandwidth allocation and sharing map is shown in Fig. 5(b). The total bandwidth is shared by ONUs based on a two-dimensional TFDM scheme. As all sub-bands are modulated on single wavelength or received by single-receiver, the peak data rate in both downstream and upstream for each ONU can be 100G. Therefore, the statistical multiplexing gain from TDM can be kept. For FDM parts, as shown in Fig. 5(b), a dedicated P2P channel can be assigned to a specific end-user by appropriate bandwidth allocation for low-latency services. Obviously, the bandwidth allocation of proposed TFDM coherent PON is more flexible as two-dimensional bandwidth allocation is enabled. However, a drawback lies in this scheme that it takes the risk of the overlap of subcarriers in the frequency domain when the lasers' wavelength of users shifted. To avoid this potential problem, there must have a proper guard band between each subcarrier, which will sacrifice the full rate.

4. Flexible deployment scenarios for C-PON

4.1. Motivations and realization

One of the significant advantages of coherent-optics is the superior sensitivity. Compared with a typical IM/DD-PON with 29 dB power budget, the C-PON based on full-field coherent detection can achieve more than 10-dB gain in sensitivity [17,41]. Even for 100G and beyond, coherent PON has been demonstrated with the power budget up to 39 dB [10]. Fig. 6 (a) shows the split ratio and fiber length for different power budget levels. It is assumed here that the power budget of IM/DD PON is 29 dB and the margin is 3 dB, while the power budget of C-PON is increased by 10 dB and the same margin is assumed. The fiber loss is 0.4-dB per km for IM/DD PON while 0.35-dB per km for C-PON, considering the loss differences of O-band (1310 nm) and C-band (1550 nm). With such an extended power budget, more flexible deployment scenarios can be supported [16].

As shown in Fig. 6(b), the power budget consists of two parts, transmission loss and split loss. Error-free transmission can be achieved as long as the total loss is below the power budget. It can be concluded as:

$$\text{PowerBudget} = \text{Loss}_T + \text{Loss}_S \quad (1)$$

With an improved power budget, it is possible to support much large

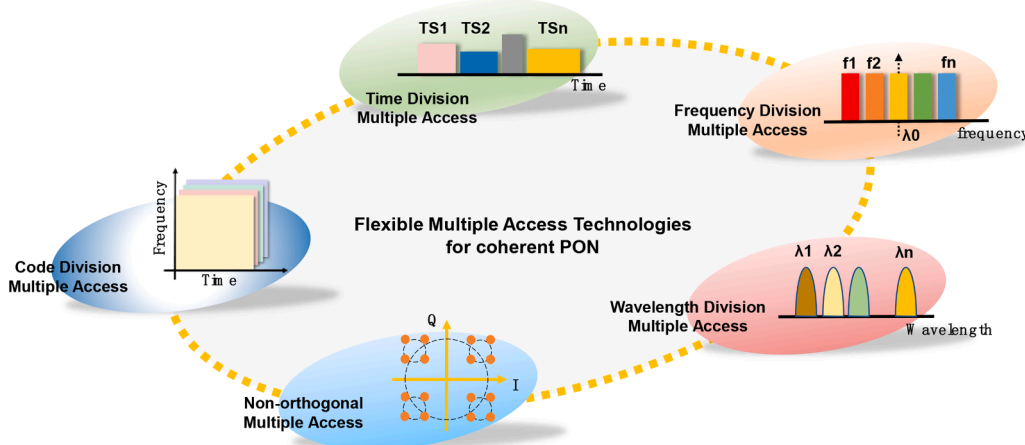


Fig. 5. Flexible multiple access technologies for coherent PON: TDMA, WDMA, FDMA, CDMA and NOMA.

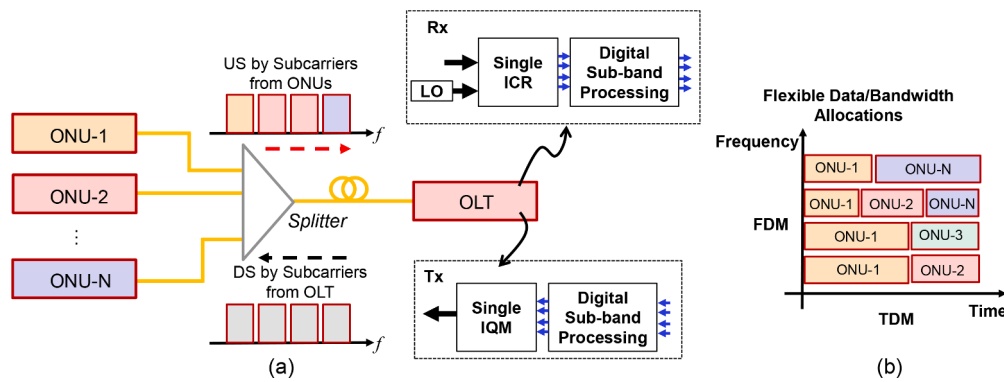


Fig. 6. (a) The architecture of TFDM coherent PON based on digital subcarrier multiplexing technology; (b) one flexible data/bandwidth allocation example.

split ratio if the fiber length is limited. For example, as shown in Fig. 6 (b), for the same fiber length of 20 km, it is possible to support 256 ONUs considering the 10-dB improved power budget. Alternatively, C-PON is also possible to support 70 km long-reach passive ODN with 8 ONUs in some extreme case. In other flexible deployment scenario, it can also support one group of users at 30-km with 32 ONUs, while continues to support 16 ONUs at 50 km in a hybrid way.

To further explain the flexibility in deployment, Fig. 7 shows the examples in flexible deployment scenarios with different fiber length and split ratios. In the PON system, the channel condition and power budget for each passive node are significantly different due to various fiber distances and the number of passive nodes. The scenarios are grouped into several areas according to different fiber transmission length and split ratios, including the urban, sub-urban and rural areas. The short reach urban areas are the dense areas that the access network should support a large number of business users. In the long reach rural areas, the business density is relatively low. And the sub-urban areas between urban areas and rural areas have the mixed reach and density. In these situations, the use of network resources is inefficient and the connection flexibility is limited by using the traditional PON architecture [16]. Fortunately, flexible C-PON is able to address this problem thanks to high data rate and power budget. For example, C-PON can cover different areas with a large variance in fiber distance and split ratios by using adaptive modulation formats or coding rates [12]. Therefore, C-PON provides more degrees of freedom for the access network in terms of deployment (See Fig. 8).

5. Discussion and summary

Recent advances of flexible C-PON in terms of access speed, architecture and deployment scenarios are reviewed and discussed in this paper. We first introduce the rate-adaptive coherent PON as a promising solution to achieve flexible C-PON. Different schemes to achieve

adaptive coherent access speed, including adaptive baud rate, adaptive modulation orders, multidimensional modulation, adaptive modulation and coding, and adaptive probabilistic shaping, are discussed. Some of these schemes are derived from the research of the IMDD-based flexible PONs. Adaptive modulation formats based on constellation probabilistic shaping is highlighted to realize fine granularity of information rate and wide flexibility. We have also discussed the flexible multiplexing schemes beyond TDM, to take the full advantages of coherent access. Multiple-access technologies, including TDM, FDM, WDM, NOMA, CDM are reviewed. A two-dimensional hybrid multiple-access based on TFDM scheme for flexible PON are introduced and highlighted. Finally, we analyze the flexible deployment scenarios under the large power budgets thanks to coherent detection.

It is worth noting that despite these flexibility advantages, many challenges remain in bringing coherent optics to the access network. For example, the cost and complexity of full field coherent transceivers are still too high for PON. The laser part is one of the critical issues for future adoption of CPON as tunable lasers are a significant part of the overall cost. Compared to the costly narrow-linewidth external cavity lasers (ECLs) used in long-haul transmission system, low-cost, small-footprint lasers with relatively large linewidth are preferred [42]. For instance, distributed feedback (DFB) lasers with larger linewidth ($\sim$ MHz linewidth) can be used allowing an acceptable power penalty for low-order modulation formats in C-PON. It can also balance the output power and total power budget for a more economical solution considering the SNR and peak data rates required for flexible applications. In addition, using a remote laser source is another way to reduce the overall cost as demonstrated in [43].

Compared to long-haul and metro coherent transmission systems which has considerable computational complexity and power consumption due to the distance-dependent DSP, the DSP in access network can be optimized in complexity and power consumption since the distance is much shorter. As analyzed in [42], many signal equalizations

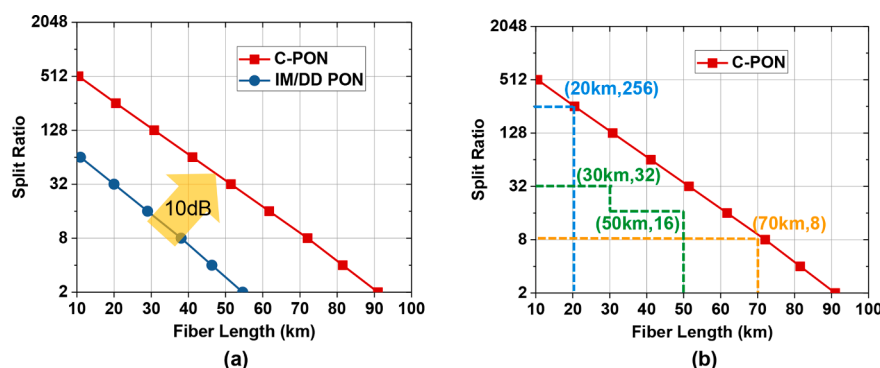


Fig. 7. The split ratio and fiber length for different PONs: (a) compare the C-PON and typical IM/DD-PON; (b) the combinations of fiber length and split ratio for C-PON.

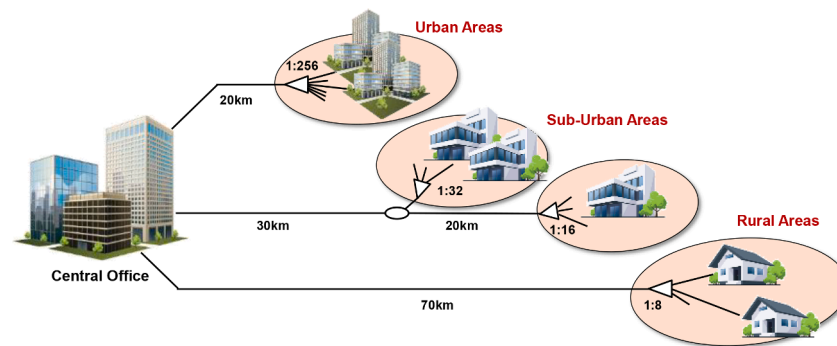


Fig. 8. Examples of C-PON with large power budget to support areas with different fiber transmission length and split ratios in flexible deployment scenarios.

can be simplified. Additionally, there is a trade-off between the DSP complexity (highly related to power consumption) and the overall performances (related to the power budget and data rates) in access network. We believe that the turning point of detection technology between IM/DD and coherent optics will be 100G access data rate, as DSP in PON demonstration of 100G and beyond based on IM/DD is becoming more and more complex. That is to say, it would be better to remain with IM/DD in access network for data rates below 100G.

The history of optical access networks reveals that the large-scale deployment and success of PON depend on the maturity of photonic and electronic technologies [44]. Coherent optics has been a great success in long-haul transmission system for over two decades. Driven by the volumes and maturation of technologies, coherent optics has entering short-reach metro and data-center markets. The success of coherent optics in short-reach metro and data-center markets will further improve the volumes and maturation of technologies. This will be beneficial for the adoption of coherent optics in access network. Additionally, by taking advantage of photonic integrated circuit (PIC) technology, coherent optics in access networks have great potential for improvement in cost, size and power consumption. We believe that the PIC technology will to be a must for future coherent optical access network instead of using separate IQ modulators and coherent receivers.

Although the use of coherent optics in next-generation PON systems offers many advantages, the main challenges of coherent PON remain and they must be resolved to promote the use of coherent optics in access networks. These challenges include the issue of high cost, complexity and power consumption for coherent optics, and upstream burst-mode in coherent detection. Some high-end users, namely business campus, or bandwidth hungry services such as mobile X-haul, would be the first applications in access network that can promote the adoption of coherent optics in access network. Therefore, a step by step adoption, progressive technology maturity and a use scenario-driven development strategy will be critical to avoid dangers or pitfalls of coherent optics in access network. There are already many efforts to simplify the coherent optics for access network [30,45]. These flexibilities discussed in this paper also applies to simplified C-PON. Therefore, we believe that the advantages of flexibility and adaptation in C-PON will become beneficial in the next generation of optical access networks.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work is partially supported by National Natural Science Foundation of China (62171137, 61925104, 62031011, 62235005), Natural Science Foundation of Shanghai (21ZR1408700), and the Major Key Project of PCL.

References

- [1] J.S. Wey, The outlook for PON standardization: a tutorial, *J. Light. Technol.* 38 (2020) 31–42.
- [2] J.S. Wey, J. Zhang, Passive Optical Networks for 5G Transport: technology and standards, *J. Light. Technol.* 37 (2019) 2830–2837.
- [3] J.A. Hernandez, R. Sanchez, I. Martin, D. Larrabeiti, Meeting the traffic requirements of residential users in the next decade with current FTTH standards: how much? How long? *IEEE Commun. Mag.* 57 (2019) 120–125.
- [4] J. Zhang, Z. Jia, Coherent passive optical networks for 100G/λ-and-beyond fiber access: recent progress and outlook, *IEEE Netw.* 8 (2022).
- [5] F. Saliou, P. Chanclou, L.A. Neto, G. Simon, J. Potet, M. Gay, L. Bramerie, H. Debregeas, Optical access network interfaces for 5G and beyond [Invited], *J. Opt. Commun. Netw.* 13 (2021) D32–D42.
- [6] R. Bonk, D. Geng, D. Khotimsky, D. Liu, X. Liu, Y. Luo, D. Nessel, V. Oksman, R. Strobel, W. Van Hoof, J.S. Wey, 50G-PON: the first ITU-T higher-speed PON system, *IEEE Commun. Mag.* 60 (2022) 48–54.
- [7] L. Xue, R. Lin, J. Van Kerrebrouck, L. Yi, J. Chen, and X. Yin, “100G PAM-4 PON with 34 dB Power Budget Using Joint Nonlinear Tomlinson-Harashima Precoding and Volterra Equalization,” in *2021 European Conference on Optical Communication (ECOC)* (2021), pp. 1–4.
- [8] J. Zhang, Z. Jia, M. Xu, H. Zhang, L.A. Campos, Efficient preamble design and digital signal processing in upstream burst-mode detection of 100G TDM coherent-PON, *J. Opt. Commun. Netw.* 13 (2021) A135–A143.
- [9] K. Matsuda, N. Suzuki, Hardware-efficient signal processing technologies for coherent PON systems, *J. Light. Technol.* 37 (2019) 1614–1620.
- [10] S. Xing, G. Li, J. Chen, J. Zhang, N. Chi, Z. He, and S. Yu, “First Demonstration of PS-QAM based Flexible Coherent PON in Burst-Mode with 300G Peak-Rate and Record Dynamic-Range and Net-Rate Product up to 7,104 dB-Gbps,” in *2022 Optical Fiber Communications Conference and Exhibition (OFC)* (2022), pp. 1–3.
- [11] F. Vaconio, O. Bertran-Pardo, Y. Pointurier, J. Fickers, A. Ghazisaeidi, G. de Valicourt, J.-C. Antona, P. Chanclou, and S. Bigo, “Flexible TDMA access optical networks enabled by burst-mode software defined coherent transponders,” in *39th European Conference and Exhibition on Optical Communication (ECOC 2013)* (2013), pp. 1–3.
- [12] M. Xu H. Zhang Z. Jia L.A. Campos Adaptive Modulation and Coding Scheme in Coherent PON for Enhanced Capacity and Rural Coverage Optical Fiber Communication Conference (OFC) 2021 2021 (OSA p. Th5I.4.
- [13] R. Borkowski, Y. Lefevre, A. Mahadevan, D. van Veen, M. Straub, R. Kaptur, B. Czerwinski, B. Cornaglia, V. Houtsmas, W. Coomans, R. Bonk, J. Maes, FLCSPON—an opportunistic 100 Gbit/s flexible PON prototype with probabilistic shaping and soft-input FEC: operator trial and ODN case studies, *J. Opt. Commun. Netw.* 14 (2022) C82–C91.
- [14] T. Pfeiffer, P. Dom, S. Bidkar, F. Fredrick, K. Christodouloupolous, R. Bonk, PON going beyond FTTH [Invited Tutorial], *J. Opt. Commun. Netw.* 14 (2022) A31–A40.
- [15] J. Zhang Z. Jia H. Zhang M. Xu J. Zhu L. Alberto Rate-Flexible Single-Wavelength TFDM 100G Coherent PON based on Digital Subcarrier Multiplexing Technology Optical Fiber Communication Conference (OFC) 2020 2020 (OSA p. W1E.5.
- [16] C. Stengrim and J. Zhang, “Coherent Optics Use Cases in Access Networks,” in *Coherent Optics for Access Networks* (CRC Press, 2019).
- [17] Z. Jia, L.A. Campos (Eds.), *Coherent Optics for Access Networks*, CRC Press, 2019.
- [18] G. Li, S. Xing, Z. Li, J. Zhang, and N. Chi, “200-Gb/s/λ Coherent TDM-PON with Wide Dynamic Range of gt;30-dB based on Local Oscillator Power Adjustment,” in *2022 Optical Fiber Communications Conference and Exhibition (OFC)* (2022), pp. 1–3.

- [19] R. Borkowski, M. Straub, Y. Ou, Y. Lefevre, Z. Jelic, W. Lanneer, N. Kaneda, A. Mahadevan, V. Huckstadt, D. van Veen, V. Houtsma, W. Coomans, R. Bonk, J. Maes, FLCS-PON – A 100 gbit/s flexible passive optical network: concepts and field trial, *J. Light. Technol.* 39 (2021) 5314–5324.
- [20] R. Zhang, N. Kaneda, Y. Lefevre, A. Mahadevan, D. van Veen, V. Houtsma, Probabilistic and Geometric Shaping for Next-Generation 100G Flexible PON, in: *In 2020 European Conference on Optical Communications (ECOC)*, (IEEE, 2020, pp. 1–4.
- [21] Cable Television Laboratories, Inc., “Data-Over-Cable Service Interface Specifications DOCSIS® 3.1,” (2014).
- [22] V.E. Houtsma, D.T. van Veen, Investigation of Modulation Schemes for Flexible Line-Rate High-Speed TDM-PON, *J. Light. Technol.* 38 (2020) 3261–3267.
- [23] J. Zhou J. He X. Lu G. Wang Y. Bo G. Liu Y. Huang L. Li H. Wang W. Mo W. Liu C. Yu Z. Li First 100Gb/s Fine-Granularity Flexible-Rate PON Based on Discrete Multi-Tone and PAPR Optimization Optical Fiber Communication Conference (OFC) 2022 2022 (Optica Publishing Group p. Th2A.23.
- [24] T. Kodama, R. Matsumoto, T. Inoue, S. Namiki, and M. Jinno, “Baud-Rate-Adaptive OLT Integrated-Coherent Transceiver for Nyquist Spectral Shaped/Channel Spaced WDM-PON,” in *2020 Opto-Electronics and Communications Conference (OECC)* (2020), pp. 1–3.
- [25] T. Kodama, R. Matsumoto, and N. Suzuki, “Demonstration of Data-rate and Power-budget Adaptive 100 Gb/s/ λ -based Coherent PON Downlink Transmission,” 3 (2017).
- [26] N. Kaneda, R. Zhang, Y. Lefevre, A. Mahadevan, D. van Veen, V. Houtsma, Experimental demonstration of flexible information rate PON beyond 100 Gb/s with probabilistic and geometric shaping, *J. Opt. Commun. Netw.* 14 (2022) A23.
- [27] A. Sheikh, A.G. i Amat, G. Liva, Probabilistic amplitude shaping with hard decision decoding and staircase codes, *J. Light. Technol.* 36 (2018) 1689–1697.
- [28] R. Matsumoto, K. Matsuda, and N. Suzuki, “Burst-Mode Coherent Detection Using Fast-Fitting Pilot Sequence for 100-Gb/sM Coherent TDM-PON System,” in *2017 European Conference on Optical Communication (ECOC)* (2017), pp. 1–3.
- [29] M.S. Erkilinc, D. Lavery, K. Shi, B.C. Thomsen, R.I. Killey, S.J. Savory, P. Bayvel, Comparison of low complexity coherent receivers for UDWDM-PONs (A-to-the-User), *J. Light. Technol.* 36 (2018) 3453–3464.
- [30] J. Li, M. Luo, Z. He, X. Xiao, and S. Yu, “Demonstration of 128×100-Gb/s Real-Time Coherent UDWDM-PON with gt;35-dB Power Budget,” in *2022 Optical Fiber Communications Conference and Exhibition (OFC)* (2022), pp. 1–3.
- [31] M. Luo, J. Li, T. Zeng, L. Meng, L. Xue, L. Yi, X. Li, Real-time coherent UDWDM-PON With dual-polarization transceivers in a field trial, *J. Opt. Commun. Netw.* 11 (2019) A166.
- [32] P. Pavon-Marino, N. Skorin-Kapov, M.V. Bueno-Delgado, J. Bäck, A. Napoli, On the benefits of point-to-multipoint coherent optics for multilayer capacity planning in ring networks with varying traffic profiles, *J. Opt. Commun. Netw.* 14 (2022) B30.
- [33] J. Back, P. Wright, J. Ambrose, A. Chase, M. Jary, F. Masoud, N. Sugden, G. Wardrop, A. Napoli, J. Pedro, M.A. Iqbal, A. Lord, D. Welch, CAPEX savings enabled by point-to-multipoint coherent pluggable optics using digital subcarrier multiplexing in metro aggregation networks, in: *in 2020 European Conference on Optical Communications (ECOC)*, (IEEE, 2020, pp. 1–4.
- [34] D. Welch, A. Napoli, J. Back, W. Sande, J. Pedro, F. Masoud, C. Fludger, T. Duthel, H. Sun, S.J. Hand, T.-K. Chiang, A. Chase, A. Mathur, T.A. Eriksson, M. Plantare, M. Olson, S. Voll, K.-T. Wu, Point-to-multipoint optical networks using coherent digital subcarriers, *J. Light. Technol.* 39 (2021) 5232–5247.
- [35] D. Izquierdo, J.A. Altabas, M. Barrio, J. Clemente, P. Millan, S. Sarmiento, J. A. Lazaro, S. Rommel, R. Puerta, J.J. Vegas-Olmos, I. Tafur Monroy, I. Garces, “Flexible resource provisioning of polarization independent coherent PONs based on non-orthogonal multiple access and multiCAP modulation,” *J. Opt. Commun. Netw.* 13 (2021) 140.
- [36] X. Wang, Y. Huang, Y. Lu, Y. Yu, and L. Li, “A Digital Coherent Optical Code Division Multiplexing Network with 16-Tb/s (2560×6.25-Gb/s) Capacity,” in *ECOC 2016; 42nd European Conference on Optical Communication* (2016), pp. 1–3.
- [37] A. Rashidinejad A. Nguyen M. Olson S. Hand D. Welch Real-Time Demonstration of 2.4Tbps (200Gbps/) Bidirectional Coherent DWDM-PON Enabled by Coherent Nyquist Subcarriers Optical Fiber Communication Conference (OFC) 2020 2020 (OSA p. W2A.30.
- [38] L. Yin, Y. Zhu, Q. Wu, H. Jiang, Q. Zhuge, W. Hu, Pilot-tone assisted successive interference cancellation for uplink power- and frequency-division multiplexing passive optical network, *J. Light. Technol.* 1–1 (2022).
- [39] M. Xu, Z. Jia, H. Zhang, L. A. Campos, and C. Knittle, “Intelligent Burst Receiving Control in 100G Coherent PON with 4×25G TFDm Upstream Transmission,” in *2022 Optical Fiber Communications Conference and Exhibition (OFC)* (2022), pp. 1–3.
- [40] Y. Zhu, B. Yang, Y. Zhong, Z. Liu, Y. Guo, J.S. Wey, X. Huang, Z. Ma, Performance Comparison of Coherent and Direct Detection Schemes for 50G PON, in: *Optical Fiber Communication Conference (OFC) 2020* (2020), Paper W1E.3, (Optica Publishing Group, 2020, p. p. W1E.3..
- [41] Z. Jia, L.A. Campos, Coherent optics ready for prime time in short-haul networks, *IEEE Netw.* 35 (2021) 8–14.
- [42] A. Shahpari, R.M. Ferreira, R.S. Luis, Z. Vujcic, F.P. Guiomar, J.D. Reis, A. L. Teixeira, Coherent access: A review, *J. Light. Technol.* 35 (2017) 1050–1058.
- [43] E. Harstead, D. Van Veen, V. Houtsma, P. Dom, Technology roadmap for time division multiplexed passive optical networks (TDM PONs), *J. Light. Technol.* 1–1 (2018).
- [44] D. Lavery and M. S. Erkilinc, “Simplified Coherent Optics for Passive Optical Networks,” in *Coherent Optics for Access Networks* (CRC Press, 2019).